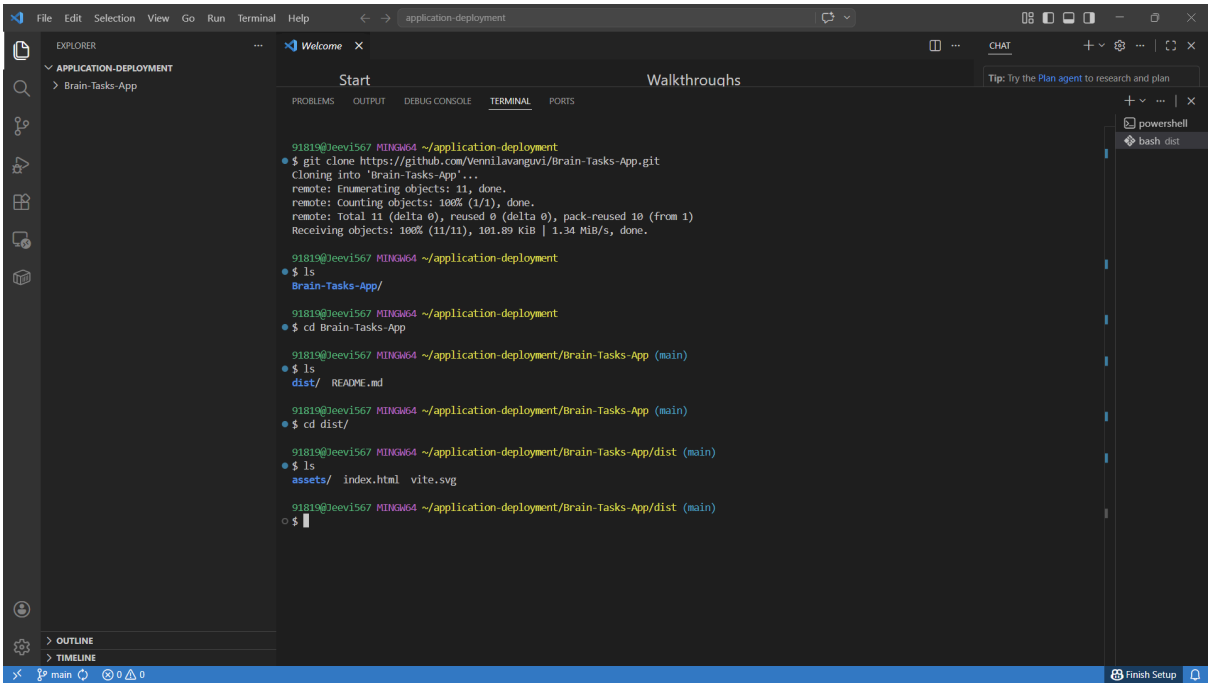
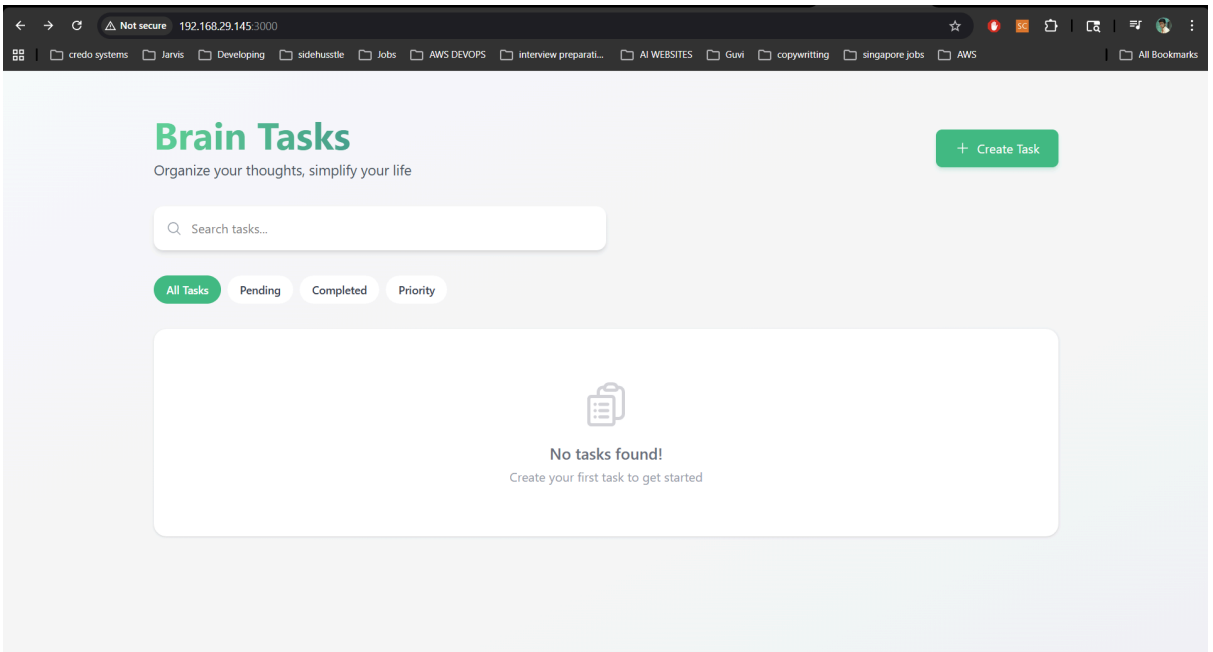
Application:

Clone the below mentioned repository and deploy the application (Run application in port 3000).
Repo URL : https://github.com/Vennilavanguvi/Brain-Tasks-App.git

I have set up in my local machine creating a folder named Application deployment where I can run this application locally.

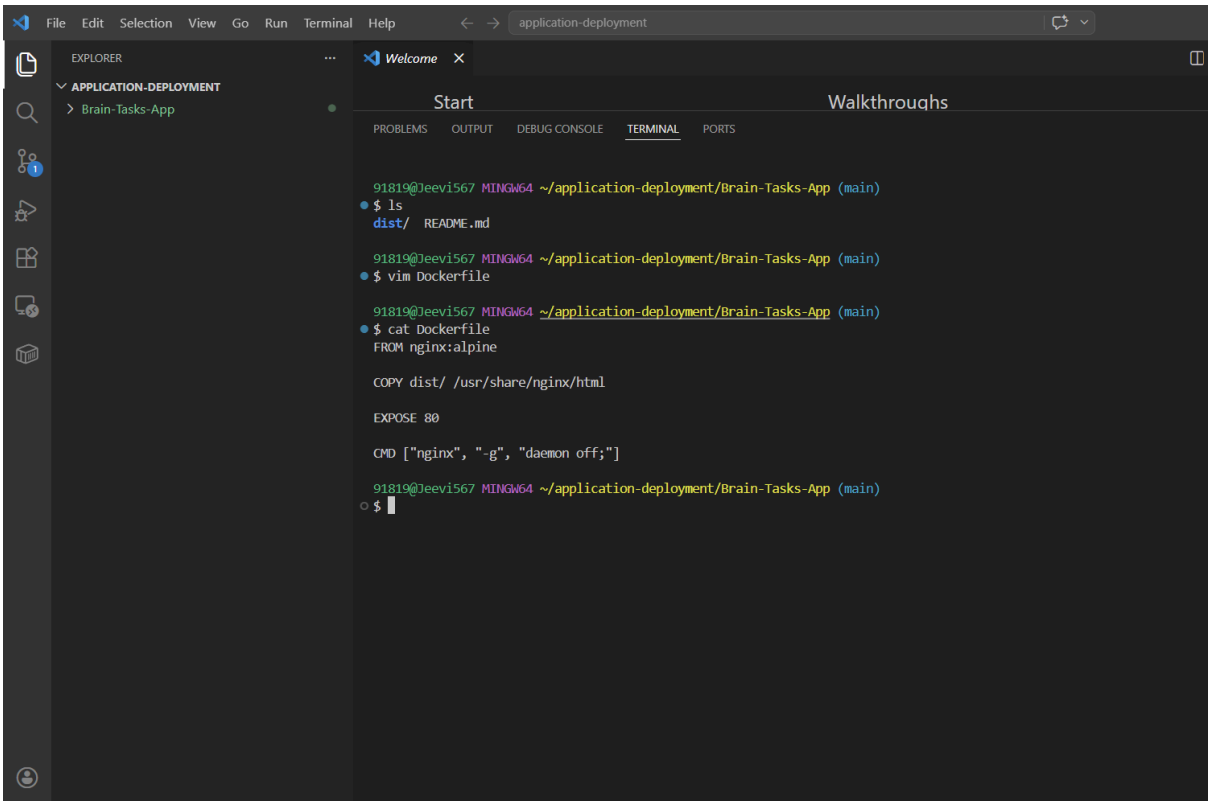


http://192.168.29.145:3000/



Docker:

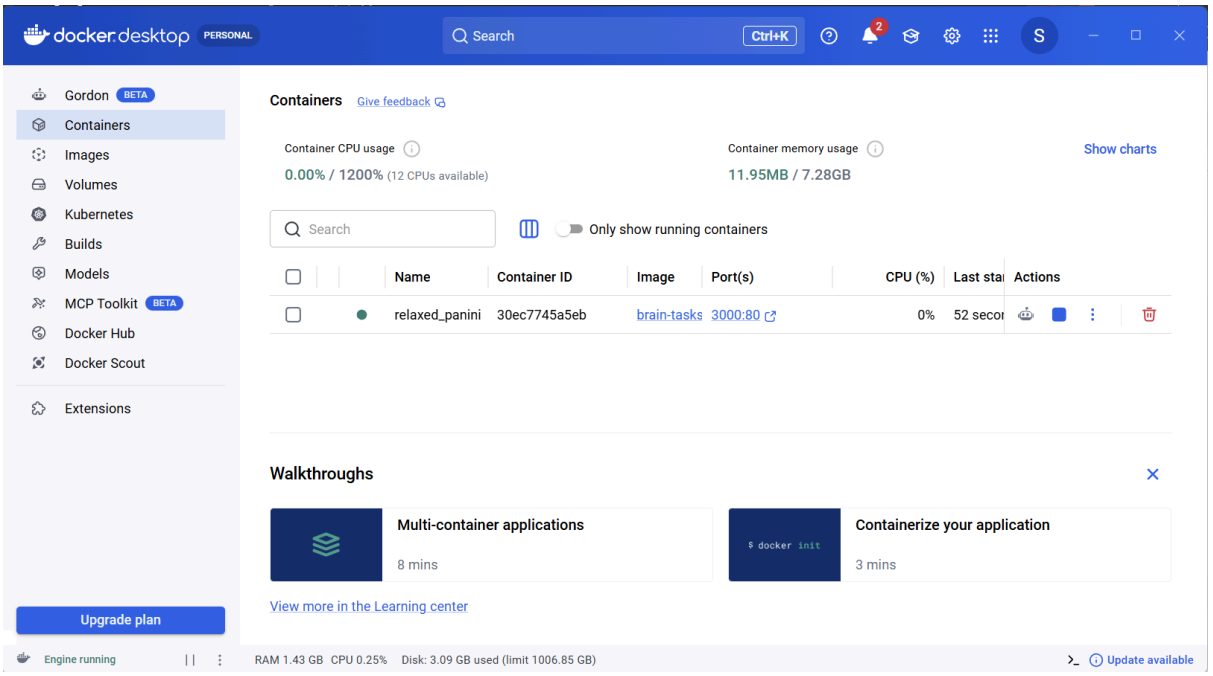
Dockerize the application by creating Dockerfile



Build an application and check output using docker image.

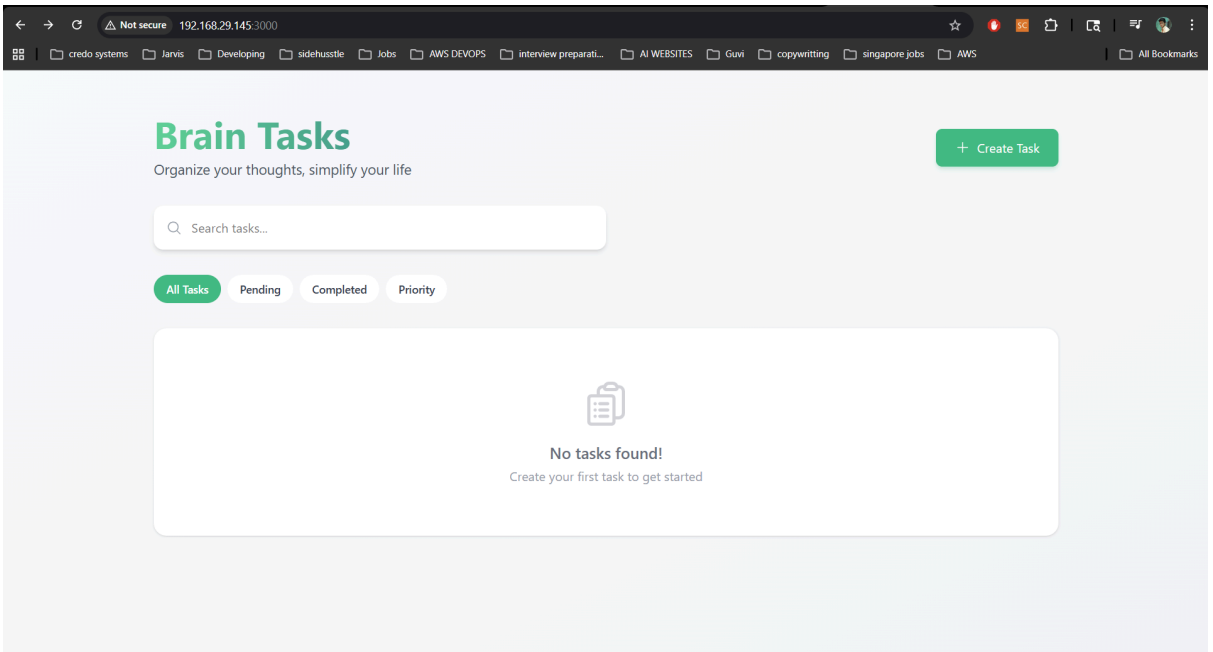
docker build -t brain-tasks-app .

docker run -p 3000:80 brain-tasks-app



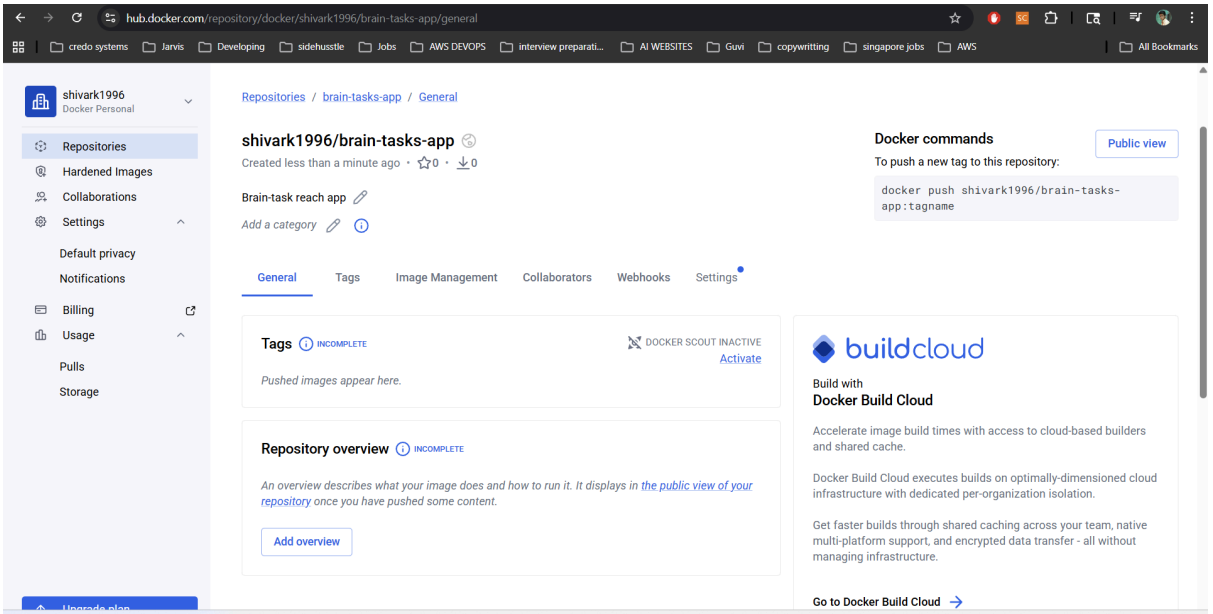
Inside container:

- App runs on **port 80**
- You map it to **3000 externally**

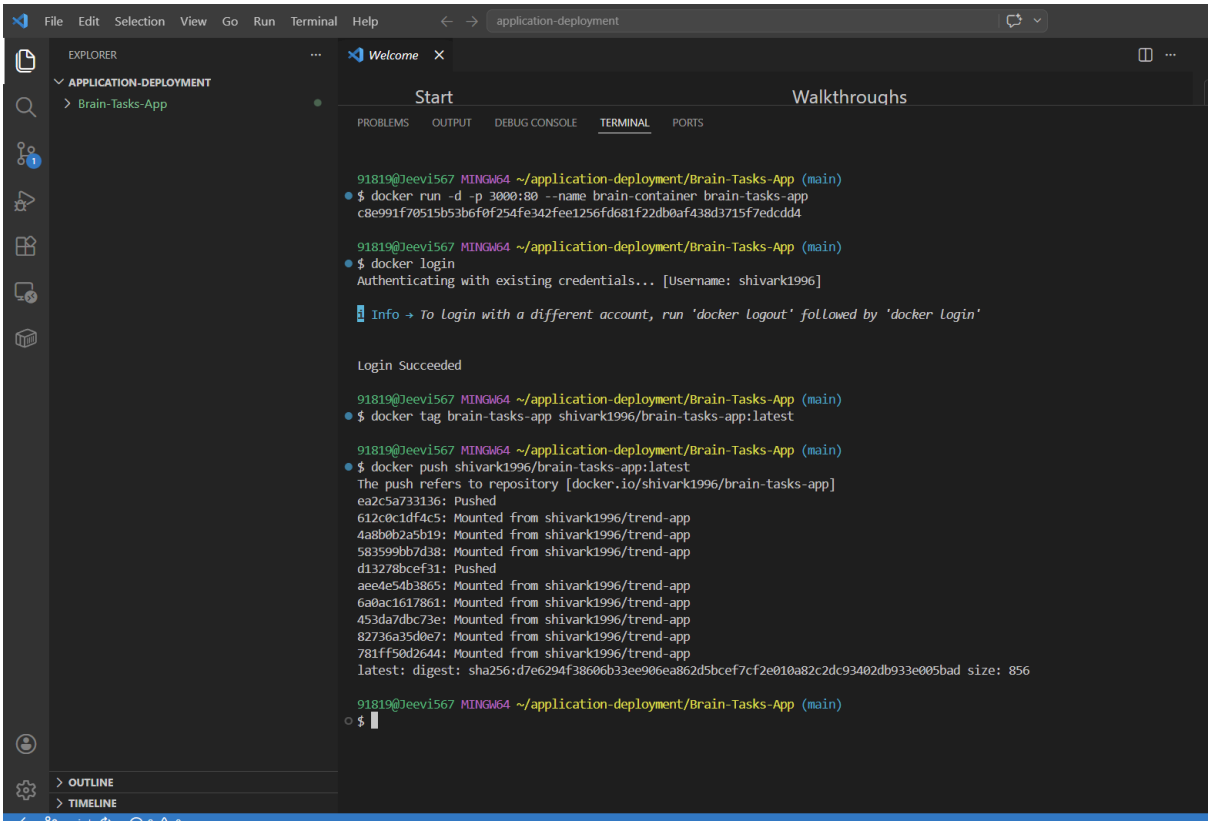


Registry:

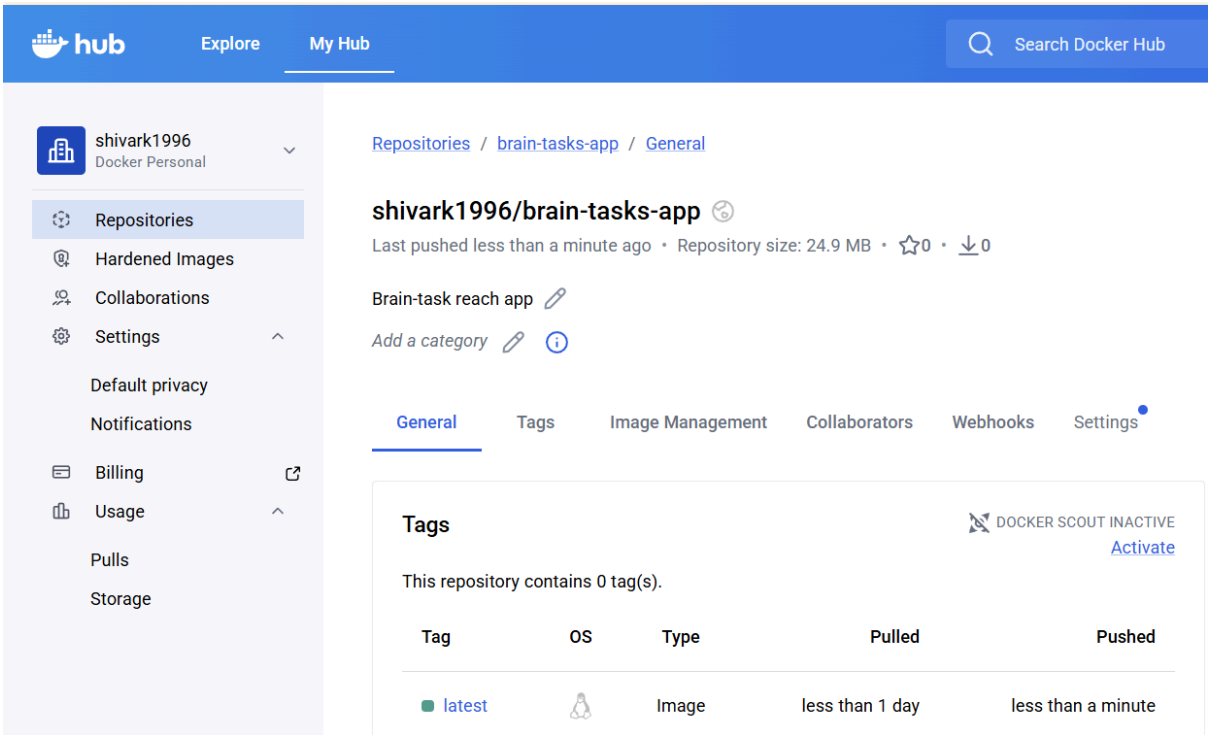
Create an AWS ECR or Dockerhub repository for storing docker images.



So I have created a repo in dockerhub named brain-task-app: and then I pushed my images to dockerhub.



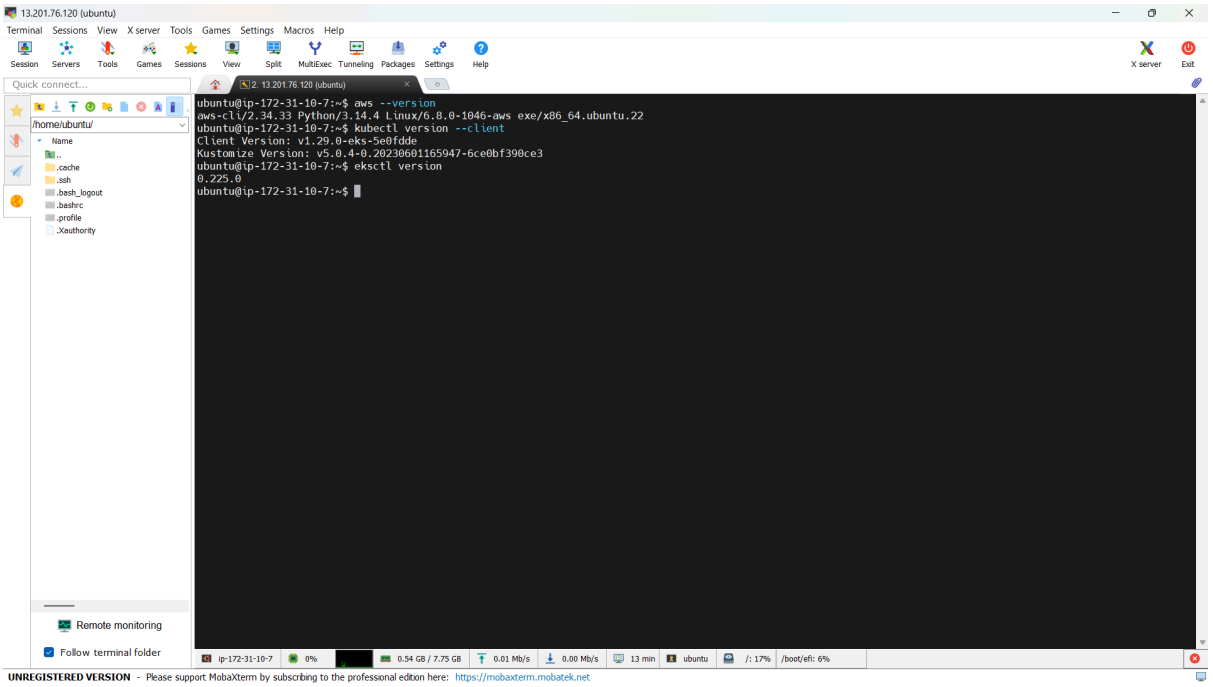
By this following command docker tag brain-tasks-app my username /brain-tasks-app:latest



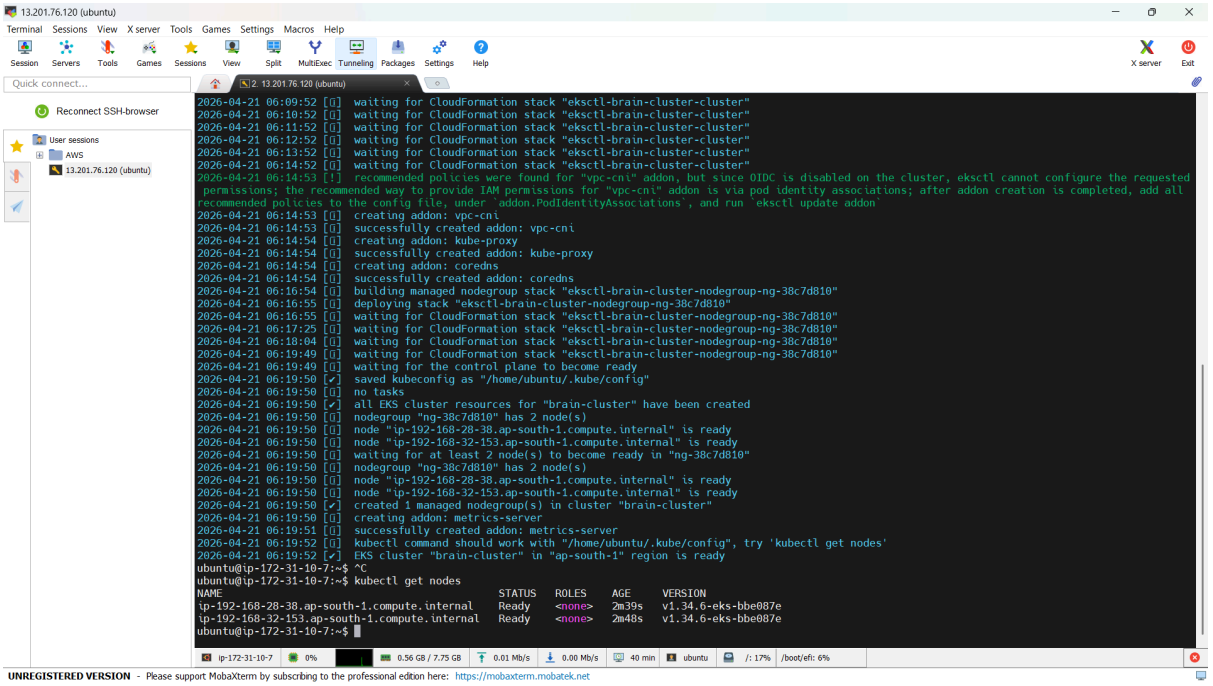
Kubernetes:

Setup Kubernetes in AWS EKS and Confirm EKS cluster is running.

For this I have created a Ec2 instance of large machine in ubuntu connected via ssh in mobaxterm



Where i Have already installed kubectl, eksctl and aws cli from the above screenshot. After this I will configure a IAM permission role and configure attach to eks management host.



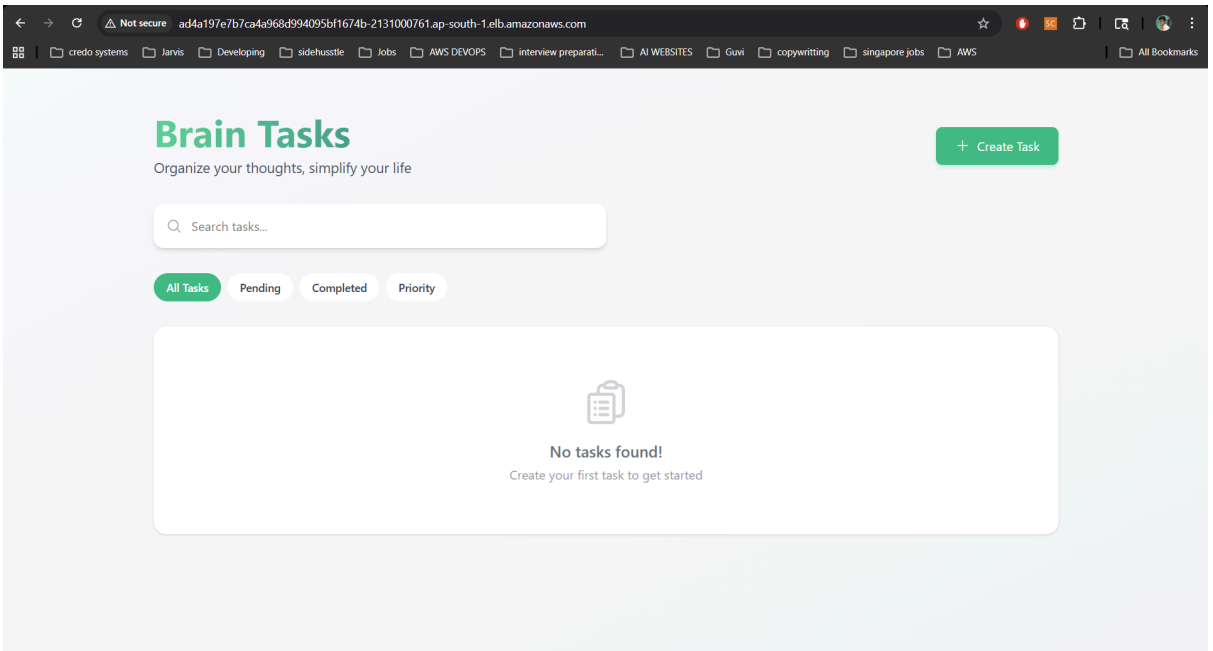
Write deployment and service YAML files.

```
ubuntu@ip-172-31-10-7:~$ vim deployment.yaml
ubuntu@ip-172-31-10-7:~$ vim service.yaml
ubuntu@ip-172-31-10-7:~$ ls
aws  awscliv2.zip  deployment.yaml  service.yaml
ubuntu@ip-172-31-10-7:~$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-app
  template:
    metadata:
      labels:
        app: brain-app
    spec:
      containers:
      - name: brain-app
        image: shivark1996/brain-tasks-app:latest
        ports:
        - containerPort: 80
ubuntu@ip-172-31-10-7:~$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-service
spec:
  type: LoadBalancer
  selector:
    app: brain-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
ubuntu@ip-172-31-10-7:~$
```

In this below screenshot I have deployed a service file to check this pods and service are working by using the following commands

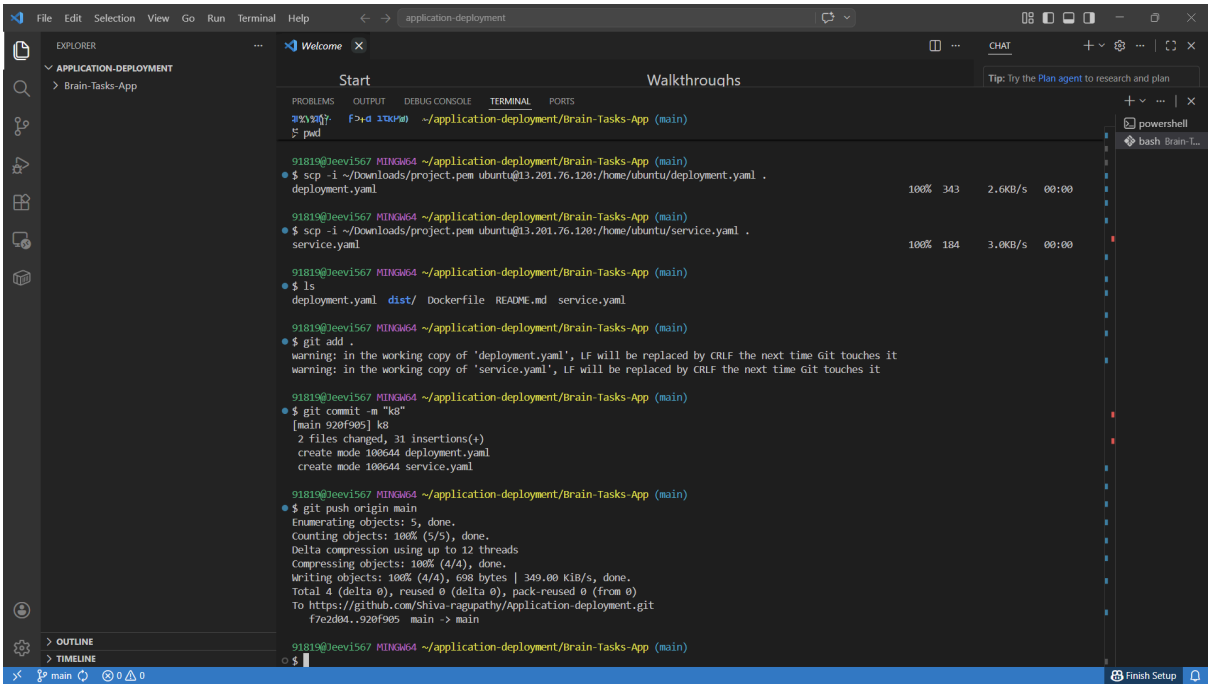
```
ubuntu@ip-172-31-10-7:~$ kubectl apply -f deployment.yaml
deployment.apps/brain-app created
ubuntu@ip-172-31-10-7:~$ kubectl apply -f service.yaml
service/brain-service created
ubuntu@ip-172-31-10-7:~$ kubectl get pods
NAME                         READY   STATUS    RESTARTS   AGE
brain-app-566dd5645c-4c6ll   1/1     Running   0          21s
brain-app-566dd5645c-f764x   1/1     Running   0          21s
ubuntu@ip-172-31-10-7:~$ kubectl get svc
NAME            TYPE           CLUSTER-IP     EXTERNAL-IP                                                              PORT(S)        AGE
brain-service   LoadBalancer   10.100.3.246   ad4a197e7b7ca4a968d994095bf1674b-2131000761.ap-south-1.elb.amazonaws.com   80:30921/TCP   20s
kubernetes      ClusterIP      10.100.0.1     <none>                                                                   443/TCP        14m
ubuntu@ip-172-31-10-7:~$ ^C
ubuntu@ip-172-31-10-7:~$ ^C
ubuntu@ip-172-31-10-7:~$
```

http://ad4a197e7b7ca4a968d994095bf1674b-2131000761.ap-south-1.elb.amazonaws.com/

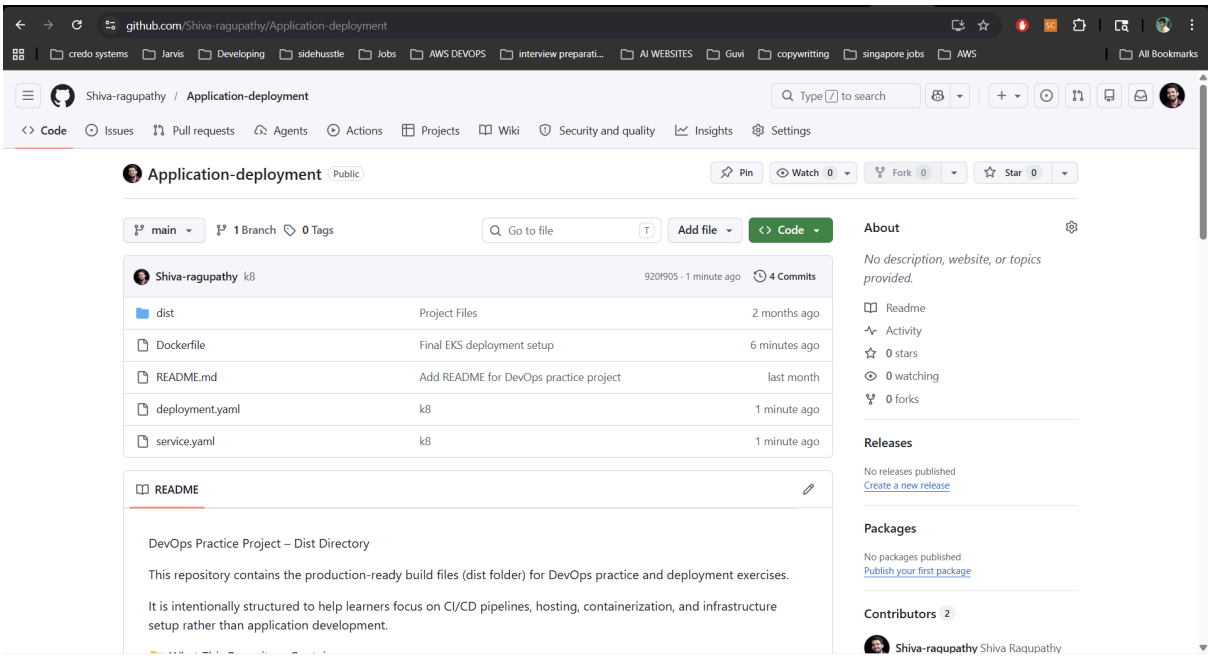


Version Control:

- Push the codebase to a Git provider (GitHub).

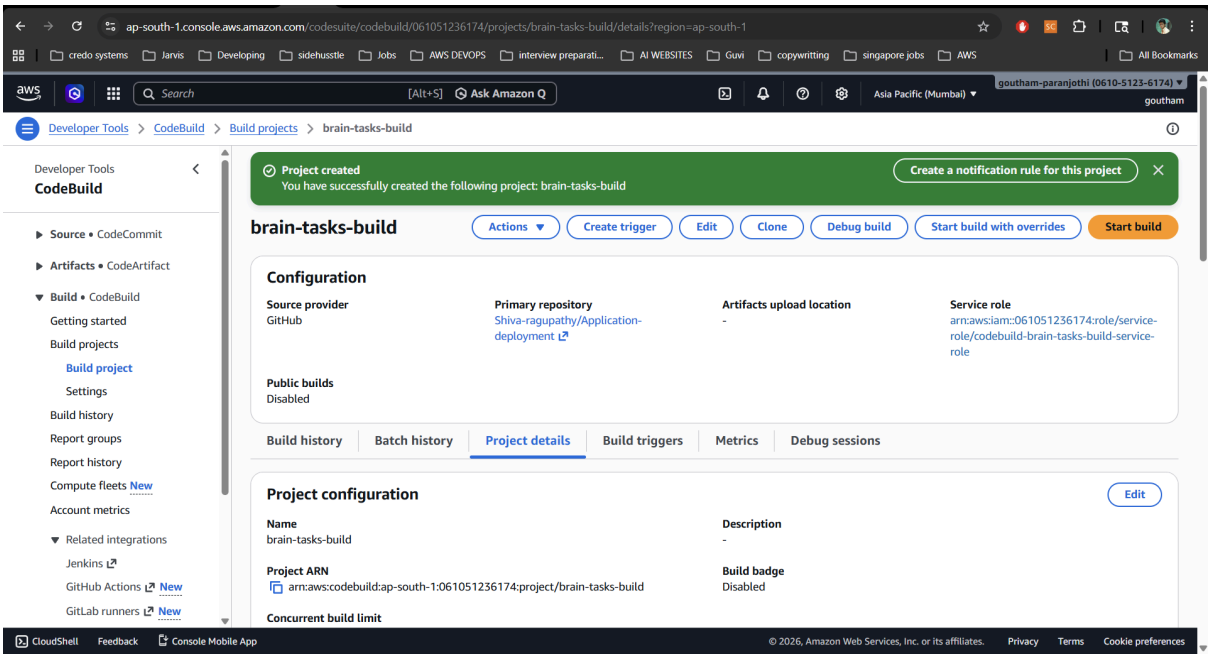


Use CLI commands to push code.

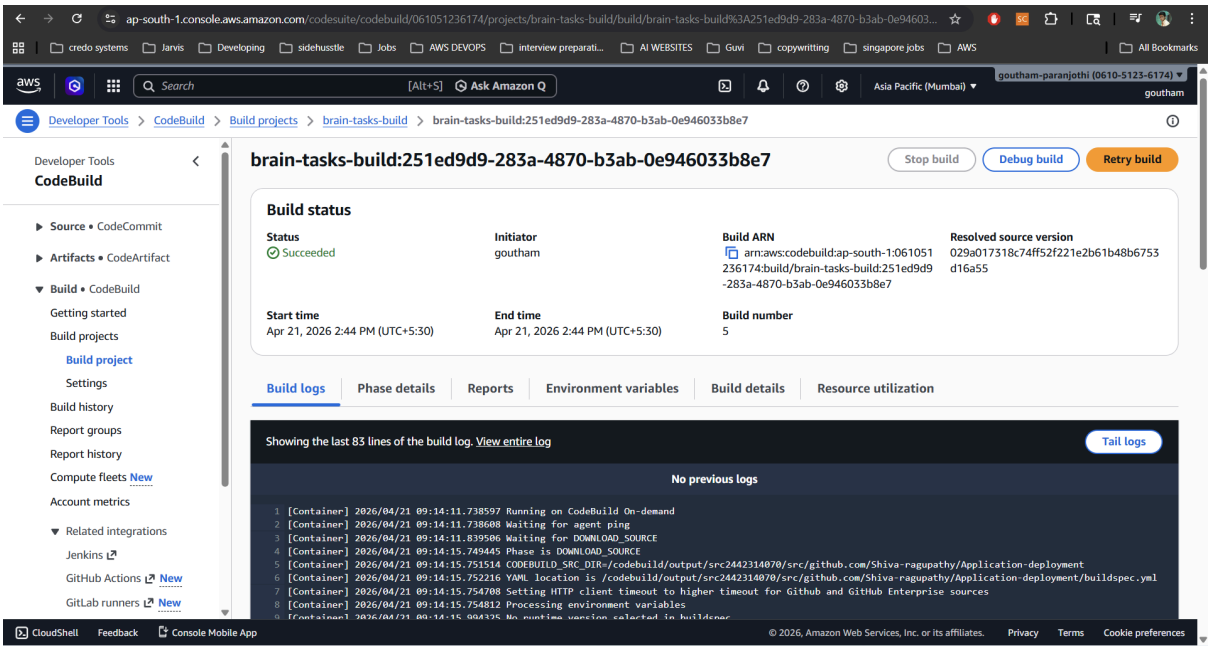


CodeBuild:

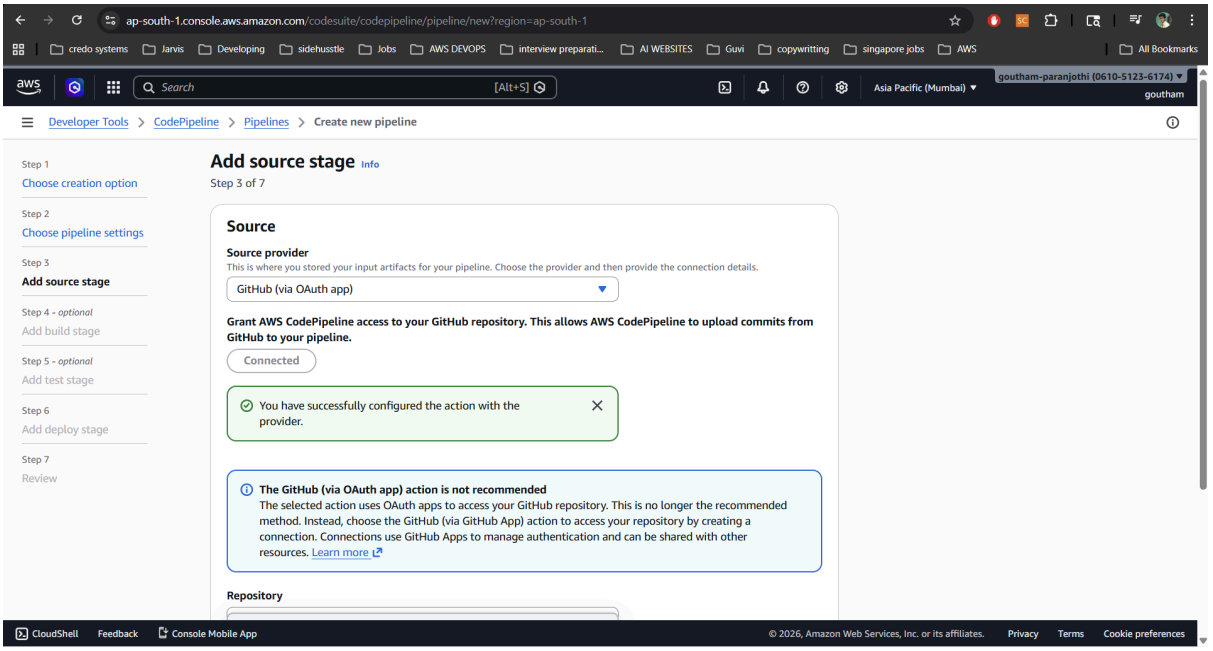
Create a CodeBuild project: Source: Connect to your repository



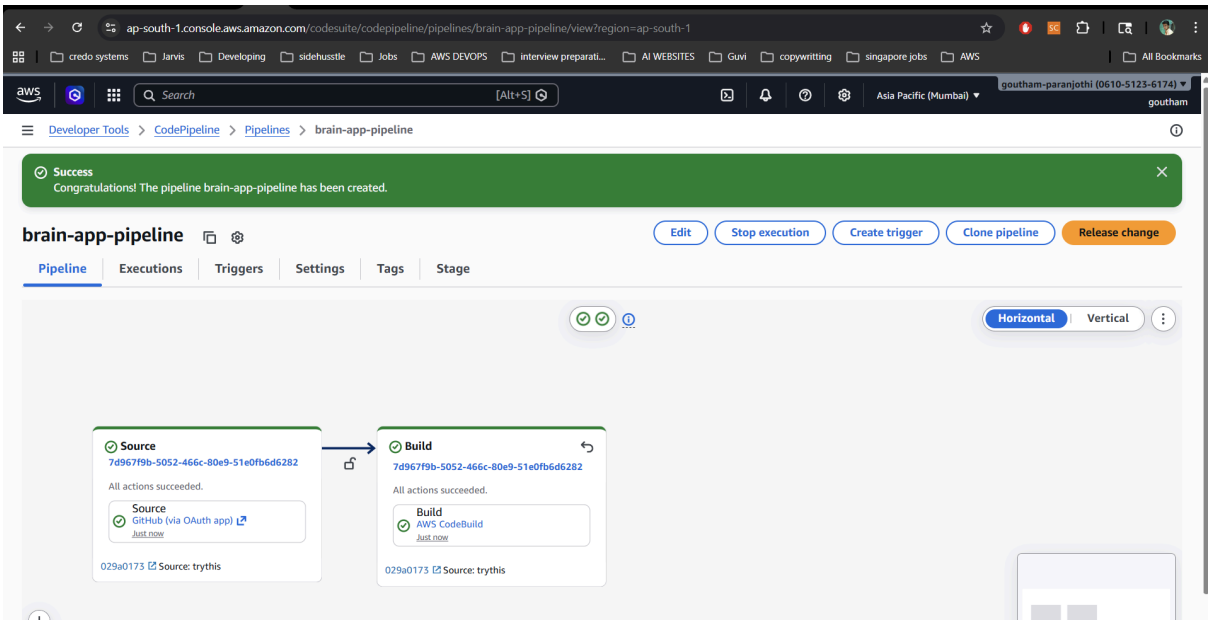
- Environment: Use managed image (Amazon Linux, Ubuntu)
- Write and define commands in buildspec.yml.



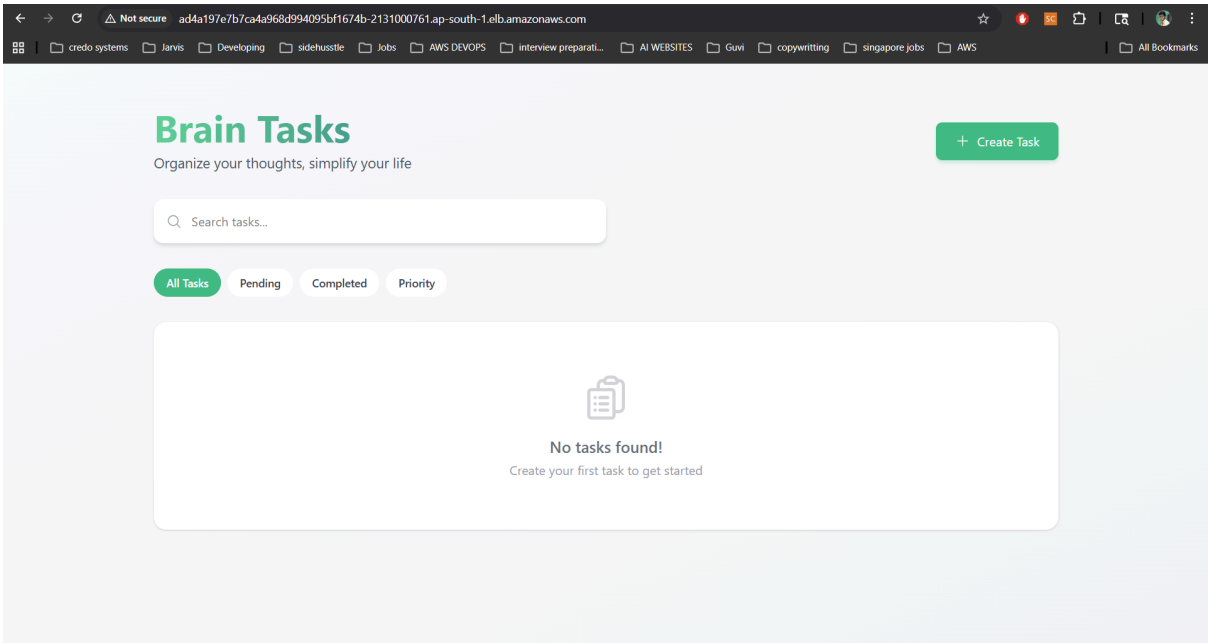
CodePipeline: Source: GitHub



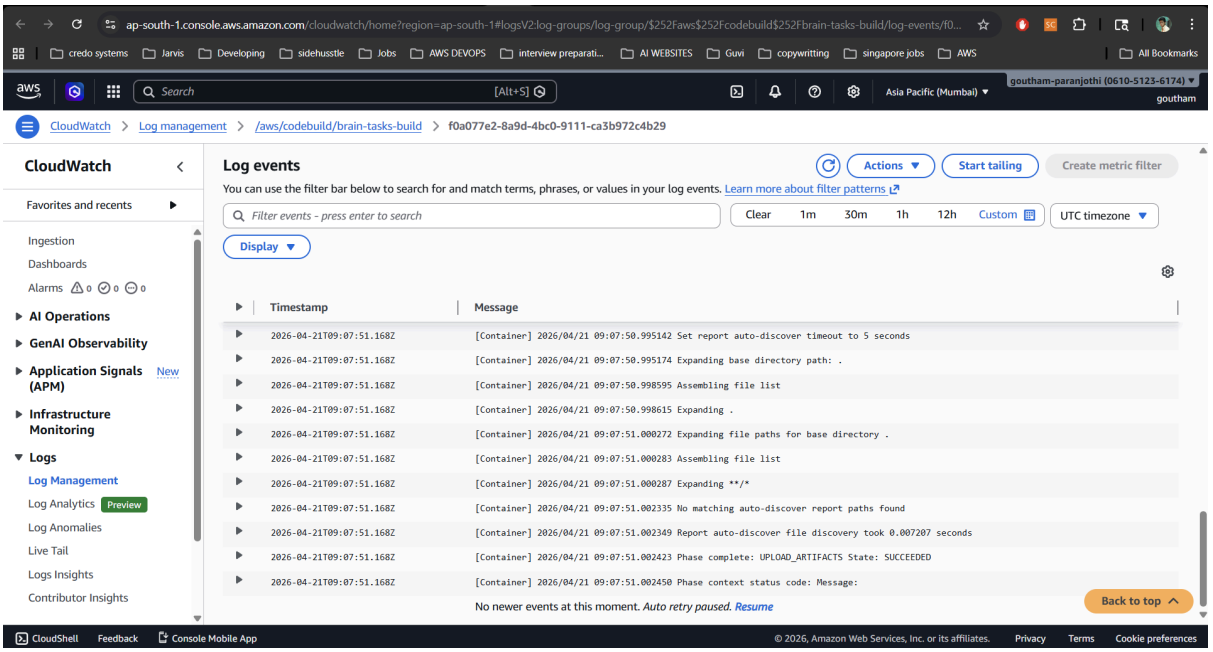
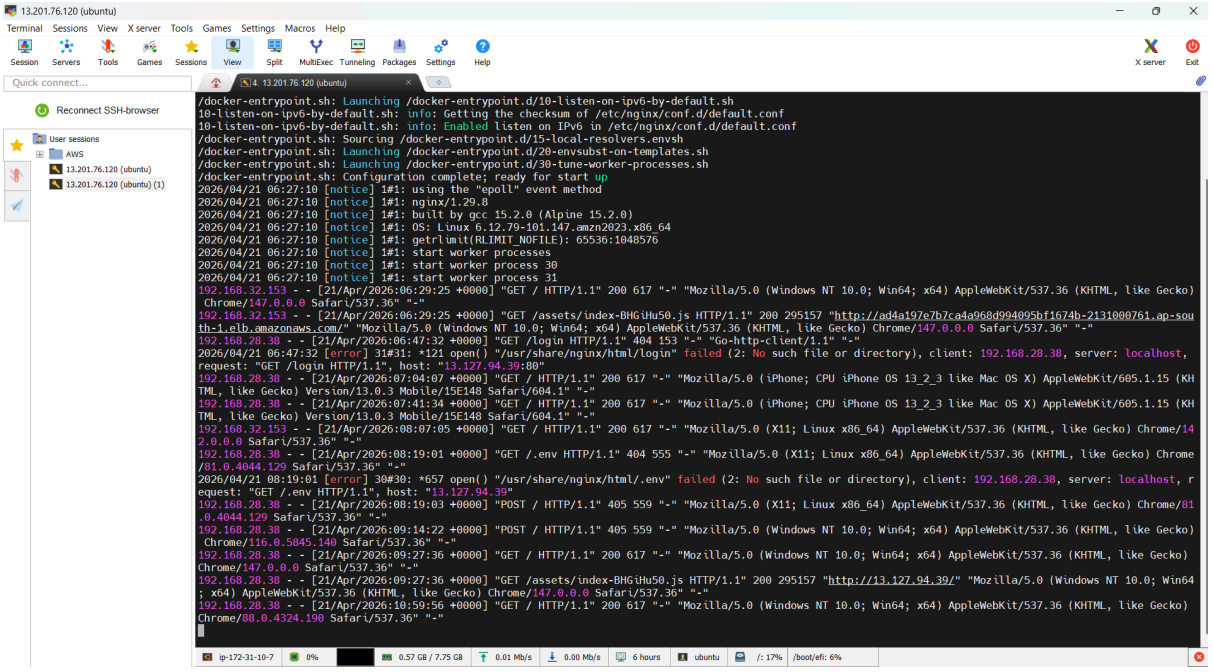
Build: AWS CodeBuild project



Deploy: Deploy to EKS via Codepipeline EKS deploy stage..



Use CloudWatch Logs to track build, deploy, and application logs.



Command kubectl logs -f brain-app-566dd5645c-4c6ll